

# Adaptive Game Difficulty Using Reinforcement Learning Based on Player Skill

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## Abstract

Teenagers and adults generally consider games to be a valuable means of fun and stress reduction. Being preoccupied in a fictional world can be relaxing and can help with mental health. But if the action and advancement of the game becomes monotonous or linear, it could cause player boredom. However, if the game is too hard for a player, frustration could be heightened and not soothed. In most digital games, the difficulty level is set beforehand by the player, for example, Easy, medium and hard. This does not consider individual skill, learning rate and playing strategy. This can make the level difficult for some players and manageable for others, which can lead to decreased engagement and satisfaction. In order to overcome this shortcoming, this research introduces an adaptive difficulty adjustment method, which adapts the difficulty level of the game dynamically based on the skill level and performance of the player. A hybrid tic-tac-toe game was created, for demonstration purposes, with a 4 x 4 grid. The computer-controlled opponent is able to analyze the player's moves, recognize game patterns and adapt its level of play. The system learns from the player by using a Q-learning algorithm and then selects a difficulty level which is appropriate for the player. It also saves each player's profile information and their acquired skill level, so that the game can be played at an appropriate level when the player returns to play a fresh game. The idea is to offer a more well-rounded and enjoyable game experience.

## 1 Introduction

In interactive digital games, it is essential to maintain an appropriate level of challenge for sustaining player engagement. If a game is consistently too difficult, players may experience frustration. On the other hand, if the level of challenge is insufficient, it can lead to boredom and reduced engagement [1]. Traditional game systems usually implement predefined difficulty levels or fixed rule-based adjustments to mitigate these issues. But adaptation to individual players is limited in this approach. A more flexible approach is adopted by Dynamic Difficulty Adjustment (DDA). Through DDA the behavior of the

game is adapted according to the performance of the player [1]. To be precise, reinforcement learning (RL) provides a promising framework for this problem because an agent can learn from repeated interactions with the game environment and adjust its decision-making based on the outcomes of previous actions. Therefore, in personalized game design, it is important to develop a lightweight mechanism that can continuously estimate player skill and adapt the strength of an RL-based opponent according to the player’s activity.

Researchers have investigated various reinforcement learning and game-theoretic approaches for decision-making in interactive environments. Early studies on off-policy temporal-difference learning established Q-learning as an effective model-free approach for learning action values, although conventional tabular methods can become difficult to scale as the state space increases [2]. Grau-Moya et al. extended soft Q-learning using Kullback–Leibler (KL) constraints and online estimation of opponent rationality. A mechanism for modeling adaptive behavior is provided, but this also introduces additional computational complexity and dependence on continuous rationality-related parameters [3]. For two-player zero-sum Markov games, Diddigi et al. proposed generalized minimax Q-learning with successive over-relaxation to improve convergence. However, the approach primarily addresses theoretical game-theoretic optimization rather than adaptation to individual human player skill [4]. Jia et al. investigated feature-based Q-learning for high-dimensional state spaces and provided theoretical sample-complexity guarantees [5], while Selvakumar and Bakolas represented pursuit-evasion problems using low-dimensional manifolds based on time-to-capture information [6]. Although these approaches improve learning or representation efficiency, their objectives differ from player-specific difficulty adaptation in interactive games. Several studies have also applied Q-learning to Tic-Tac-Toe and related board-game environments. Jiang and Wang used tabular Q-learning with self-play over 50,000 episodes [7], while Krasnikov investigated the effects of learning-rate and discount-factor configurations through systematic experiments [8]. Ho et al. combined optimistic initialization with decaying exploration over 300,000 episodes and reported approximately a 90% win/tie rate [9]. These approaches demonstrate the effectiveness of Q-learning for learning game strategies, but their primary objective is to improve playing strength against fixed or random opponents rather than to regulate the strength of the agent according to a human player’s demonstrated ability. Gannavarapu investigated reinforcement-learning-based DDA in turn-based combat by sampling actions from ranked Q-values, but reported performance limitations related to disk I/O and update delays [1]. Lin et al. proposed a CPU-efficient Deep Principal Component Analysis Network (RQDNN) for visual arcade games, focusing on game-score maximization rather than personalized difficulty adaptation [10]. Overall, existing approaches demonstrate the effectiveness of reinforcement learning for game-playing and adaptive decision-making, but several gaps remain. Many methods focus on maximizing agent performance, require comparatively complex learning or representation mechanisms, or do not incorporate a lightweight mechanism for persistent, player-specific skill estimation and difficulty adaptation.

Motivated by these limitations, this work proposes a Hybrid Adaptive Reinforcement Learning Framework for player-specific difficulty adjustment in a  $4 \times 4$  Tic-Tac-Toe environment. The proposed system combines Tabular Q-learning with  $\epsilon$ -greedy exploration and a rule-based tactical policy that prioritizes immediate winning and blocking opportunities. The system also estimates player expertise from recent gameplay outcomes and maps the estimated skill to five difficulty levels: Very Easy, Easy, Normal, Hard, and Expert. Rather than directly modifying the learned Q-table, the difficulty mechanism controls the agent’s effective exploration rate and the probability of applying tactical behavior. Specifically, the effective exploration rate is calculated as

$$\epsilon_{\text{effective}} = \min(1, \epsilon \times F_d),$$

where  $F_d$  is a difficulty-dependent factor. The player expertise score is maintained within the range  $0 \leq \text{Skill} \leq 100$ , with human wins, draws, and RL-agent wins contributing differently to the recent performance score. The system further maintains persistent player profiles containing learning and performance information, allowing the learned Q-values and adaptive parameters to be restored across multiple sessions. Since gameplay interactions directly generate the learning experience, the proposed approach does not require a pre-collected training dataset. Through the combination of online Q-learning, tactical decision-making, player expertise estimation, and persistent DDA, the proposed framework aims to provide a computationally lightweight approach for adapting the strength of an RL-based game opponent to the demonstrated skill of individual players.

## 1.1 Our Contribution

The major contributions of this work are summarized as follows:

- Developed an adaptive  $4 \times 4$  Tic-Tac-Toe game integrated with a Tabular Q-Learning agent.
- Implemented an online reinforcement learning approach, eliminating the requirement for a pre-existing training dataset.
- Designed a hybrid decision-making approach by combining Tabular Q-Learning,  $\epsilon$ -greedy exploration, and rule-based tactical strategies.
- Introduced a player expertise estimation mechanism based on the player’s recent gameplay performance.
- Developed a Dynamic Difficulty Adjustment (DDA) mechanism that automatically adapts the game difficulty from Very Easy to Expert according to the estimated player expertise.

## 2 Material and Methods

The initial plan was to make a complex arcade game to do the whole research work. But being inexperienced in devloping game we have made a simple tic tac toe game using pythons pygame library to demonstrate the work flow of our whole process. A methodological diagram 1 is provided to represent the overall workflow of our entire system.

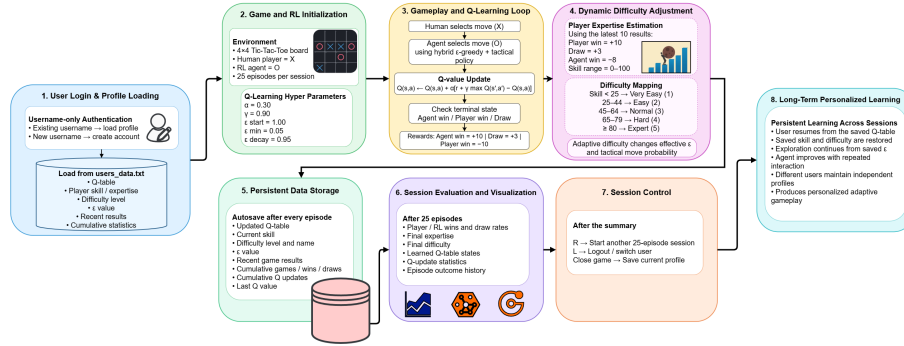


Figure 1: Methodology flow

We used Pygame library for making the game in Python which is fully functional 4x4 Tic-Tac-Toe game with an interactive graphical user interface. The interface was made to be easy and intuitive to interact with the game and watch their actions in several episodes.

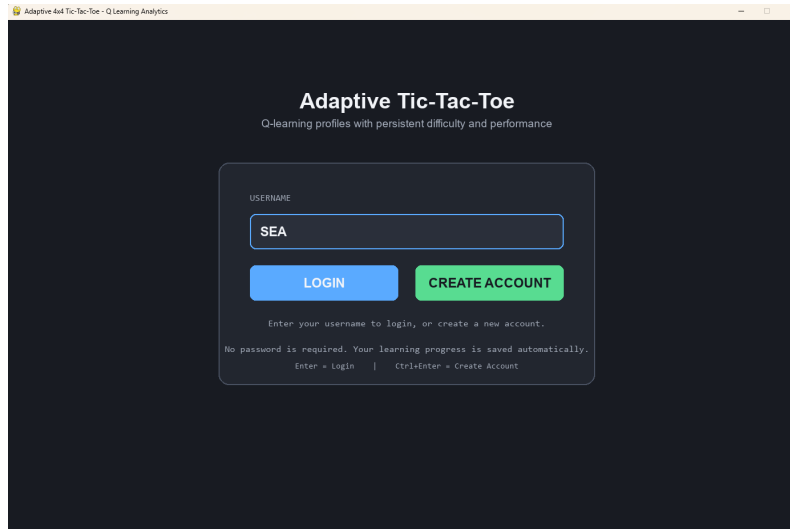


Figure 2: Login Interface

We added a player profile system in addition to the game environment itself, so that it could be used as a base for personalized and adaptive gaming. Players are able to maintain their own profile with the system recording relevant information about the difficulty level the player is likely to have to play at and their level of skill. This enables the game to differentiate between each player and keep track of each player’s progress. The learned profile information is then used by the adaptive difficulty mechanism to tweak the behaviour and skill level of the agent based on player’s performance. Therefore, this first step sets the core game environment, and also provides the base needed to implement adaptive gameplay for each player.

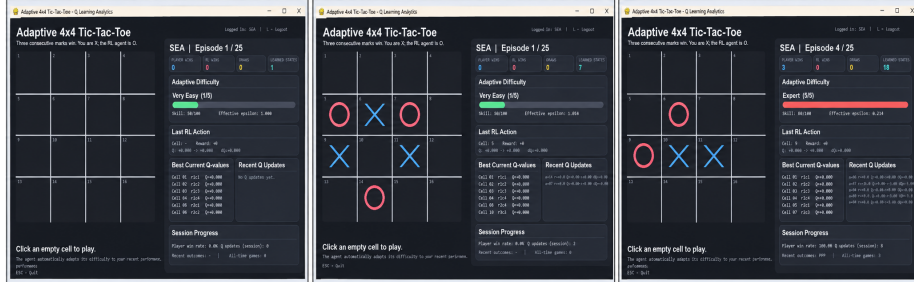


Figure 3: Game Interface

The interface portrayed in fig 3 is developed to contain the game environment of  $4 \times 4$  Tic-Tac-Toe in addition to a separate Reinforcement Learning (RL) monitoring panel to observe the agent gameplay and the learning behavior of the agent at the same time. The game interface contains the interactive game board for human player to compete against the RL agent. It also shows key game data, such as the episode being played, game results, player profile, estimated skill and dynamically assigned game difficulty.

The proposed system’s main learning process is taking place in the game loop. There are 25 episodes in each session (1 full Tic-Tac-Toe game equals 1 episode). The human player plays using the symbol “X”, while the reinforcement learning agent uses “O”. In each episode, the agent continuously interacts with the environment of the game, chooses actions, and gets rewarded or not and updates its Q values.

Initially, the human player selects an available cell on the  $4 \times 4$  game board. After the player’s move, the RL agent observes the current board configuration, which represents the current state  $s$ . The agent then selects one of the available cells as its action  $a$ . Action selection is performed using a hybrid strategy that combines an  $\epsilon$ -greedy policy with tactical decision-making. The  $\epsilon$ -greedy mechanism allows the agent to balance exploration and exploitation, whereas the tactical component enables the agent to consider important moves such as winning opportunities and blocking the opponent.

After an action is performed, the Q-value associated with the corresponding state-action pair is updated using the standard Q-learning update rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[ r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (1)$$

where  $Q(s, a)$  represents the current Q-value of action  $a$  in state  $s$ ,  $\alpha$  is the learning rate,  $r$  is the received reward,  $\gamma$  is the discount factor, and  $\max_{a'} Q(s', a')$  represents the maximum expected future Q-value from the next state  $s'$ . In the proposed system, the learning rate is set to  $\alpha = 0.30$  and the discount factor is set to  $\gamma = 0.90$ .

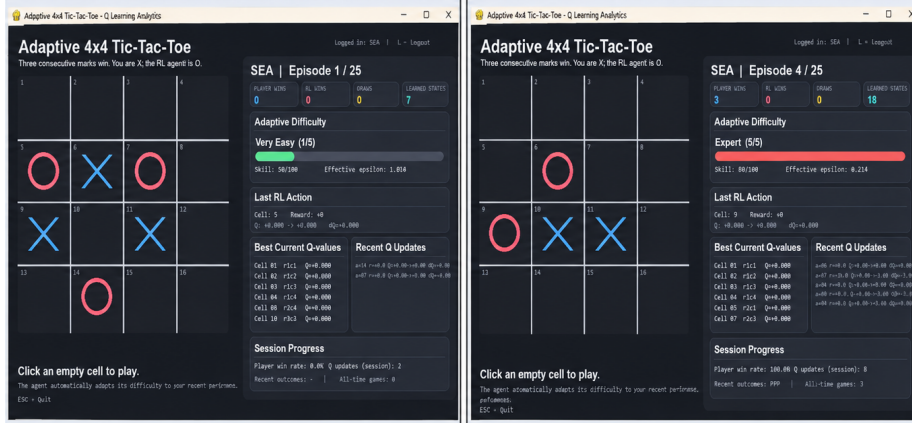


Figure 4: Early and Late Stage

After each turn, the game state is evaluated and if a terminal state is met, the game terminates. There are three possible outcome scenarios: agent wins, human player wins and draw. Reward Mechanism: RL is rewarded by +10 if he wins, +3 if he draws, and -10 if the human wins. If the state is not a terminal state, then the immediate reward is still 0. Such rewards steer the agent in direction that will make it more likely to obtain desirable outcomes.

After an episode is completed, the outcome is noted and the player's stats are adjusted which can be seen in 4. The system then re-calculates the player's level of expertise and changes the level of difficulty in the game. The exploration rate  $\epsilon$  is updated as well based on the learning process. Relevant information, including Q-value changes, episode results, player expertise, difficulty level, and other performance statistics, is recorded for subsequent evaluation and persistent storage.

This interaction process is repeated until all 25 episodes of the current session are completed. Therefore, the game play loop enables the agent to learn incrementally from repeated interaction while simultaneously providing the performance information required by the dynamic difficulty adaptation mechanism.

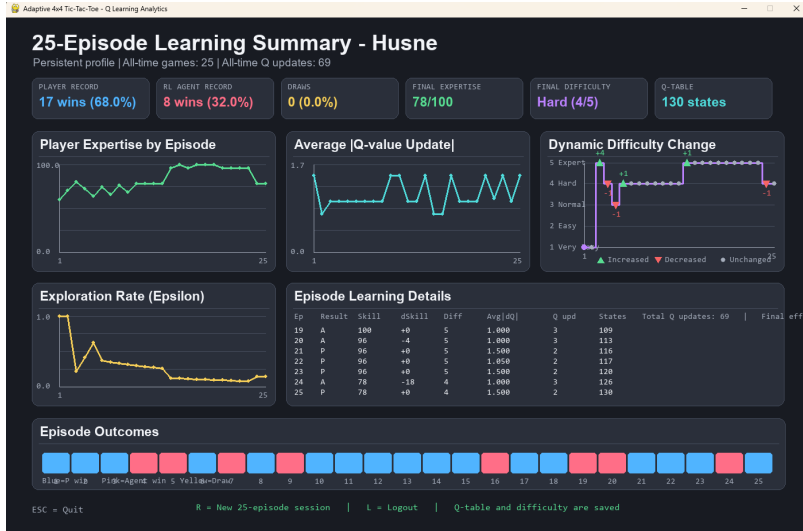


Figure 5: Post Episodic Summary

## 2.1 Dataset Description

There is no existing dataset used traditionally for training the proposed system. Instead, the data are collected directly from each user while they play the game. The system records the user’s performance in games, including results and interactions, in the game playing process. The system records the user’s performance in the game playing process, including the result of the game and interactions with the system. Meanwhile, the Q-values are updated after each move as a result of the current state, the action taken and the reward received. This enables the reinforcement learning agent to slowly improve by playing the game.

The data gathered by the user are also employed to gauge the skill level of the player and modify the level of the game. A separate summary report is created every 25 episodes. The report displays the overall performance, Q-value changes, skill advancement, difficulty changes and learning progress of the player. The system can then learn and adapt to each user without the need for a training set.

## 2.2 Tools / Models / Algorithms Used

The proposed system employs a hybrid approach that combines Tabular Q-Learning, an  $\epsilon$ -greedy action-selection strategy, and a rule-based adaptive difficulty mechanism. Q-learning enables the agent to improve its decision-making through interaction with the player, whereas the rule-based component provides tactical game knowledge and controls the difficulty of the agent according to the estimated expertise of the human player.

### 2.2.1 Tabular Q-Learning

Tabular Q-learning is used as the primary reinforcement learning algorithm. Each configuration of the  $4 \times 4$  Tic-Tac-Toe board represents a state  $s$ , while selecting an available board position represents an action  $a$ . The Q-table stores the expected utility of performing each possible action from a particular state.

After the agent interacts with the environment, the corresponding Q-value is updated according to:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[ r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (2)$$

where  $\alpha = 0.30$  represents the learning rate and  $\gamma = 0.90$  represents the discount factor. The reward  $r$  is determined by the outcome of the game. The agent receives +10 for winning, +3 for a draw, and  $-10$  when the human player wins. Non-terminal transitions receive a reward of zero.

Through repeated episodes, the Q-values are updated and the agent gradually learns which actions are more beneficial for different board states.

### 2.2.2 $\epsilon$ -Greedy Action Selection

An  $\epsilon$ -greedy strategy is used to balance exploration and exploitation. During exploration, the agent may select an alternative available action rather than relying entirely on its currently learned Q-values. During exploitation, the agent favors actions having higher Q-values.

The exploration rate is initially set to:

$$\epsilon_0 = 1.0, \quad (3)$$

and is gradually reduced using:

$$\epsilon_{t+1} = \max(\epsilon_{\min}, \epsilon_t \times 0.95), \quad (4)$$

where  $\epsilon_{\min} = 0.05$ . Consequently, the agent explores more during the initial learning stage and increasingly utilizes its learned knowledge as training progresses.

### 2.2.3 Rule-Based Tactical Decision Making

In addition to Q-learning, a rule-based tactical mechanism is incorporated into the agent's action-selection process. This prevents the agent from depending exclusively on an initially sparse Q-table and allows it to make reasonable game decisions even during the early stages of learning. The tactical rules examine the current board state and prioritize actions according to immediate game conditions. The main rules are:

1. If the agent has an immediate winning move, select the winning cell.
2. If the human player has an immediate winning move, block that cell.

3. Otherwise, evaluate available positions for useful tactical opportunities.
4. Prefer strategically useful positions, including central positions, when appropriate.
5. If no tactical rule is selected, use the Q-learning-based action selection process.

The probability of applying tactical behavior is also controlled by the current difficulty level. Therefore, easier difficulty levels intentionally use tactical knowledge less frequently, whereas harder levels use it more frequently.

#### 2.2.4 Rule-Based Player Expertise Estimation

The player’s expertise is estimated from recent gameplay performance. The system considers the most recent game outcomes and assigns performance scores according to:

Table 1: Player Expertise Score Based on Game Outcome

| Game Outcome      | Expertise Score |
|-------------------|-----------------|
| Human Player Wins | +10             |
| Game is a Draw    | +3              |
| RL Agent Wins     | −8              |

The system uses the recent performance history to calculate the player’s current expertise score, which is bounded within the range:

$$0 \leq Skill \leq 100. \quad (5)$$

As a result, consistent player victories increase the estimated expertise, while repeated losses reduce it.

#### 2.2.5 Rule-Based Dynamic Difficulty Adjustment

A rule-based Dynamic Difficulty Adjustment (DDA) mechanism maps the estimated player expertise to one of five difficulty levels. The rules used by the system are:

Table 2: Player Skill-Based Difficulty Levels

| Skill Range          | Difficulty Level |
|----------------------|------------------|
| $Skill < 25$         | Very Easy (1)    |
| $25 \leq Skill < 45$ | Easy (2)         |
| $45 \leq Skill < 65$ | Normal (3)       |
| $65 \leq Skill < 80$ | Hard (4)         |
| $Skill \geq 80$      | Expert (5)       |

During the initial games of a new player, the system starts from the Very Easy difficulty level to obtain an initial estimate of the player’s ability. Difficulty does not directly modify the learned Q-table. Instead, it changes how the agent uses its learned and tactical knowledge. In particular, the system modifies the effective exploration rate and the probability of tactical decision-making according to the selected difficulty level.

The effective exploration rate is computed as:

$$\epsilon_{\text{effective}} = \min(1, \epsilon \times F_d), \quad (6)$$

where  $F_d$  is a difficulty-dependent factor. The implemented factors are:

| Difficulty | $F_d$ | Tactical Probability |
|------------|-------|----------------------|
| Very Easy  | 1.35  | 0.05                 |
| Easy       | 1.10  | 0.15                 |
| Normal     | 0.80  | 0.35                 |
| Hard       | 0.50  | 0.70                 |
| Expert     | 0.25  | 0.92                 |

Thus, at lower difficulty levels, the agent behaves more exploratively and uses tactical decisions less frequently. As the player’s expertise increases, the effective exploration is reduced and the probability of using strong tactical moves increases. This produces an opponent whose behavior dynamically adapts to the demonstrated ability of the player.

### 2.2.6 Persistent User-Based Learning

The Q-learning process is persistent across multiple sessions. After each episode, the user’s learning and performance information is stored in the user profile. This includes the Q-table, expertise score, difficulty level, exploration rate, recent results, and cumulative performance statistics.

When an existing user logs in again, the previously stored Q-table and adaptive parameters are restored instead of initializing the learning process from the beginning. Therefore, the system provides both **long-term reinforcement learning** and **user-specific dynamic difficulty adaptation**.

Overall, the proposed method can be described as a **Hybrid Adaptive Reinforcement Learning Framework**, where Tabular Q-Learning provides experience-based learning, the  $\epsilon$ -greedy policy balances exploration and exploitation, tactical rules provide immediate game intelligence, and rule-based DDA controls the strength of the agent according to the player’s observed performance.

## 2.3 Performance Evaluation

The proposed system does not use a traditional pre-collected dataset; therefore, a conventional training, validation, and testing split is not applicable. Instead, the reinforcement learning agent follows an **online learning approach**, where

training data are generated dynamically through interactions between the human player and the game environment. Each state, action, reward, and resulting state obtained during gameplay contributes to the learning process.

The performance of the Q-learning agent is evaluated based on the changes in its learned Q-values and gameplay outcomes. The Q-value is updated using the temporal-difference (TD) error:

$$\delta = r + \gamma \max_{a'} Q(s', a') - Q(s, a) \quad (7)$$

where  $\delta$  represents the temporal-difference error,  $r$  represents the immediate reward,  $\gamma$  is the discount factor,  $s$  represents the current state,  $a$  represents the selected action, and  $s'$  represents the resulting state.

The Q-learning update is then performed as:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \delta \quad (8)$$

where  $\alpha = 0.30$  represents the learning rate and  $\gamma = 0.90$  represents the discount factor. The magnitude of the Q-value change,  $|\Delta Q|$ , is recorded to observe the learning behavior of the agent.

### 3 Experimental Analysis

After every 25 episodes, a summary of the entire iteration is generated to evaluate the learning and adaptation behavior of the proposed system. The summary presents the overall gameplay outcomes, Q-value updates, player expertise progression, and changes in the dynamically adjusted difficulty level. Graphical representations are also generated to visualize the Q-value learning progress and difficulty adaptation throughout the episodes. These results provide a clear understanding of how effectively the RL agent learns from user interactions and adapts its behavior according to the player's performance.

The average Q-value change represents the mean magnitude of all Q-value updates performed during a particular episode. It indicates the overall amount of learning occurring within that episode. A larger  $\text{Avg}|\Delta Q|$  indicates that the agent is making relatively larger adjustments to its learned Q-values, whereas a smaller value indicates comparatively smaller updates.

$$\text{Avg}|\Delta Q| = \frac{1}{N} \sum_{i=1}^N |\Delta Q_i|$$

where  $N$  is the total number of Q-value updates performed in an episode.

**Maximum Q-Value Change ( $\text{Max}|\Delta Q|$ ):** The maximum Q-value change represents the largest absolute Q-value update observed during a particular episode. It identifies the strongest individual learning adjustment made by the agent within that episode. A high  $\text{Max}|\Delta Q|$  generally occurs when an action receives a strong reward or penalty, causing a larger update to the corresponding Q-value.

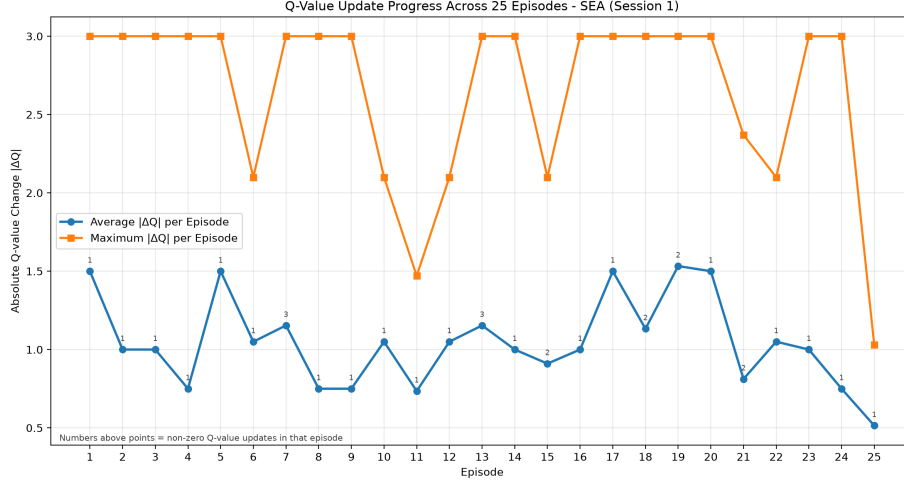


Figure 6: Update of Q values

$$\text{Max}|\Delta Q| = \max_{1 \leq i \leq N} |\Delta Q_i|$$

As shown in the Q-value update graph 6, both  $\text{Avg}|\Delta Q|$  and  $\text{Max}|\Delta Q|$  fluctuate across the 25 episodes. This indicates that the magnitude of learning updates varies according to the states, actions, and rewards experienced by the RL agent during each game.

The dynamic difficulty adaptation graph 7 illustrates how the difficulty level changes across the 25 gameplay episodes according to the player's estimated expertise. The system uses five difficulty levels: Very Easy (1), Easy (2), Normal (3), Hard (4), and Expert (5). The player's skill is continuously recalculated from recent gameplay performance, and the corresponding difficulty level is adjusted automatically.

As observed in the graph, the difficulty initially changes between Expert, Hard, and Normal as the system evaluates the player's recent performance. For example, a skill value of 60 corresponds to the Normal level, while a skill value of 78 corresponds to Hard. As the player's performance improves, the difficulty subsequently increases to Expert. From approximately Episode 17 onward, the difficulty remains at Expert, indicating that the player's estimated expertise remains within the highest difficulty range. The dynamic difficulty level is determined according to the player's expertise score ( $S$ ), as presented in Table 2.

**Github Repository :** [https://github.com/Syed-Eftasum-Alam/Adaptive\\_Tic-Tac-Toe-Game](https://github.com/Syed-Eftasum-Alam/Adaptive_Tic-Tac-Toe-Game)

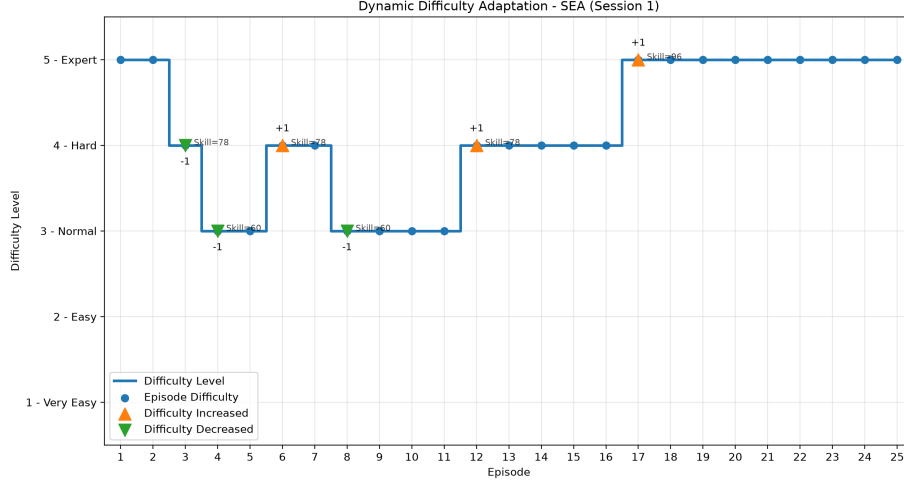


Figure 7: Dynamically change of difficulty

### 3.1 Ablation Study and Experimental Analysis

To evaluate the contribution of the major components of the proposed Adaptive Tic-Tac-Toe agent, an ablation study was conducted by selectively removing the Q-learning component, tactical policy, and dynamic difficulty adaptation (DDA). Four configurations were evaluated: (i) *Full*, containing Q-learning, tactical decision rules, epsilon-greedy exploration, and DDA; (ii) *No Tactical*, in which the tactical decision layer was removed; (iii) *No Adaptive*, in which dynamic difficulty adaptation was disabled; and (iv) *No Q*, in which Q-learning was removed and the agent relied on the rule-based tactical policy.

Unlike the preliminary experiments, which evaluated the agents directly during a short interactive training session, the final ablation experiment used a controlled training-evaluation protocol. Each configuration was trained for 800 games and subsequently evaluated for 200 additional games with learning disabled. Five independent random seeds were used for each model-opponent combination. Three automated opponents were considered: *Random*, which selects legal moves randomly; *Heuristic*, which prioritizes immediate winning moves and blocking moves; and *Strong Tactical*, which additionally incorporates center preference and therefore represents a substantially stronger deterministic opponent. The resulting experiment contains  $4 \times 3 \times 5 \times 1000 = 60,000$  games.

During the 200-game evaluation phase, Q-value updates and exploration were disabled. Therefore, the reported evaluation performance measures the policies obtained after training rather than continued online learning. Agent win rate, draw rate, player win rate, average reward, final player skill, difficulty level, and the number of learned states were recorded. Mean values and standard deviations were calculated over the five random seeds.

The results in Table 3 demonstrate that the tactical component is the most

Table 3: Frozen-evaluation results over five random seeds. Values are mean percentages over 200 evaluation games per seed. Standard deviations are reported in parentheses.

| Model       | Opponent        | Agent Win (%) | Draw (%)   | Player Win (%) | Reward  | Final Skill  | Final Difficulty |
|-------------|-----------------|---------------|------------|----------------|---------|--------------|------------------|
| Full        | Random          | 95.2 (1.40)   | 0.1 (0.22) | 4.7 (1.25)     | 9.053   | 0.0          | 1.0              |
| Full        | Heuristic       | 55.7 (0.45)   | 0.4 (0.42) | 43.9 (0.65)    | 1.192   | 52.2 (14.87) | 2.8 (0.84)       |
| Full        | Strong Tactical | 0.0 (0.00)    | 0.0 (0.00) | 100.0 (0.00)   | -10.000 | 100.0        | 5.0              |
| No Tactical | Random          | 40.7 (2.75)   | 0.0 (0.00) | 59.3 (2.75)    | -1.860  | 93.6 (7.40)  | 5.0              |
| No Tactical | Heuristic       | 1.4 (0.55)    | 0.0 (0.00) | 98.6 (0.55)    | -9.720  | 98.4 (3.58)  | 5.0              |
| No Tactical | Strong Tactical | 0.0 (0.00)    | 0.0 (0.00) | 100.0 (0.00)   | -10.000 | 100.0        | 5.0              |
| No Adaptive | Random          | 95.2 (1.40)   | 0.1 (0.22) | 4.7 (1.25)     | 9.053   | 50.0         | 1.0              |
| No Adaptive | Heuristic       | 38.2 (2.51)   | 0.3 (0.67) | 61.5 (2.98)    | -2.321  | 50.0         | 1.0              |
| No Adaptive | Strong Tactical | 0.0 (0.00)    | 0.0 (0.00) | 100.0 (0.00)   | -10.000 | 50.0         | 1.0              |
| No Q        | Random          | 98.5 (0.50)   | 0.0 (0.00) | 1.5 (0.50)     | 9.700   | 0.0          | 1.0              |
| No Q        | Heuristic       | 83.6 (2.82)   | 0.0 (0.00) | 16.4 (2.82)    | 6.720   | 3.2 (5.02)   | 1.0              |
| No Q        | Strong Tactical | 0.0 (0.00)    | 0.0 (0.00) | 100.0 (0.00)   | -10.000 | 100.0        | 5.0              |

important component for short-horizon game-playing performance. Removing tactical decision making causes the agent win rate against the Heuristic opponent to fall from 55.7% in the Full configuration to only 1.4%. The corresponding average reward also decreases from 1.192 to -9.720. This indicates that Q-learning alone, without explicit recognition of immediate winning and blocking opportunities, is insufficient to reliably handle the tactical structure of the game within the available training budget.

A similar effect is observed against the Random opponent. The No Tactical configuration achieves only a 40.7% win rate, whereas the Full model achieves 95.2%. The substantial difference confirms that the tactical layer provides an effective inductive bias by explicitly handling short-horizon threats that may require considerably more experience for tabular Q-learning to discover.

Interestingly, the No Q configuration achieves the highest performance against both the Random and Heuristic opponents, with win rates of 98.5% and 83.6%, respectively. This configuration uses the tactical rules without Q-learning and therefore demonstrates that the handcrafted immediate-win, blocking, center-preference, and strategic-position rules are highly effective for this relatively small board. Consequently, Q-learning should not be interpreted as independently improving raw short-term win rate in this experiment. Instead, its value lies in learning state-action preferences beyond the explicitly encoded tactical rules. The result also highlights an important limitation of evaluating a hybrid RL system solely according to raw game-winning percentage: a strong rule-based policy can outperform a learning component when the environment is small and the tactical structure is easy to encode.

The effect of dynamic difficulty adaptation is most clearly visible against the Heuristic opponent. The Full model achieves a 55.7% agent win rate, compared with 38.2% for the No Adaptive configuration. More importantly, the Full model dynamically changes its difficulty according to the estimated player skill, whereas the No Adaptive configuration remains at a fixed difficulty level. For the Full configuration against the Heuristic opponent, the mean final skill is 52.2 and the mean final difficulty is approximately 2.8, compared with a fixed difficulty level of 1 for the No Adaptive configuration. The Full model therefore does not simply maximize its own winning probability; it modifies its behavior

in response to the evolving estimated player skill.

The skill progression further illustrates this behavior. Across the five seeds against the Heuristic opponent, the Full configuration ended with skill values of 54, 43, 68, 32, and 64, respectively. These differences demonstrate that the adaptive mechanism responds to the outcome history rather than forcing every run toward the same fixed difficulty. In contrast, the No Adaptive configuration retains a constant skill estimate of 50 and a constant difficulty level of 1.

The Strong Tactical opponent produced a 0% agent win rate for all four configurations. All games against this opponent ended in player victories, and the average reward was  $-10.0$ . This result should not be interpreted as evidence that all configurations are equally capable. Rather, it indicates that the selected Strong Tactical opponent is considerably more difficult than the other automated opponents and creates a saturated evaluation condition in which the four configurations cannot distinguish their relative performance. In addition, the agent acts as the second player in the game, making the result particularly challenging under a strong tactical policy. Thus, the Strong Tactical condition is retained as a stress test rather than as the primary discriminative ablation condition.

The learned-state statistics also support the distinction between the learning and rule-based configurations. The Full model learned, on average, approximately 1761 states against the Heuristic opponent, while the No Q configuration learned no Q states because Q-learning was removed. The No Tactical model still learned approximately 1250 states, but its poor evaluation performance demonstrates that state learning alone does not guarantee effective tactical behavior. The combination of learned state-action values with explicit tactical safeguards is therefore more appropriate for the present environment than relying exclusively on either component.

Overall, the ablation results indicate that the proposed system benefits from a hybrid architecture. The tactical policy provides strong immediate game competence, Q-learning provides additional state-dependent learning, and DDA modifies the agent’s effective challenge level according to the estimated player skill. The results also show that these components serve different objectives: tactical rules primarily improve decision quality, Q-learning supports adaptation through experience, and DDA targets the interaction balance rather than maximizing the agent’s raw win rate.

### 3.2 Benchmark Analysis

To contextualize the performance of the proposed Adaptive Tic-Tac-Toe system, its results were compared with five representative studies covering reinforcement learning for Tic-Tac-Toe and reinforcement learning-based dynamic difficulty adjustment (DDA). Because the selected studies employ different environments, board sizes, player roles, opponent policies, training durations, and evaluation protocols, the reported values are treated as *contextual benchmarks* rather than direct state-of-the-art comparisons.

The most directly comparable studies are the Tic-Tac-Toe experiments of

Table 4: Contextual comparison with representative reinforcement learning and dynamic difficulty adjustment studies.

| Study                           | Environment                        | Approach                                                                 | Evaluation                                                            | Reported Result                                                                                                                                                                                 |
|---------------------------------|------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Konen and Bartz-Beielstein [11] | 3×3 Tic-Tac-Toe                    | TD learning with function approximation                                  | 40 selected states and 300-game tournament against Random and Minimax | RL vs. Random: 100% X wins. RL vs. Minimax: 100% ties. The best agent was trained using 10,000 self-play games.                                                                                 |
| Silva et al. [12]               | DotA/MOBA                          | Dynamic difficulty adjustment using performance-based opponent switching | Agent-vs-agent and agent-vs-user experiments                          | Adaptive behavior maintained balanced matches in approximately 85% of one condition; other scenarios reported approximately 90% compatible performance and 80% balance against a hard opponent. |
| Climent et al. [13]             | Single-player action game          | RL-based DDA with two Q-tables                                           | Computer-driven and human-player evaluation                           | The framework targets an approximately 50% winning/losing ratio and personalized challenges according to player skill.                                                                          |
| de Oliveira and Chaimowicz [14] | Fighting game                      | RL-based opponent with skill-matching reward and win-rate penalty        | Generalization to different opponent behaviors                        | The proposed reward constrains opponent strength while regularization improves adaptability and generalization to unseen opponents.                                                             |
| Timóteo et al. [15]             | 3×3 Tic-Tac-Toe with robotic arm   | Q-learning, Random, and MiniMax                                          | Training against multiple predefined opponents                        | Against Random, Q-learning achieved 77.38% victory as X and 32.73% as O. Against MiniMax, victory was 0%, with 52.19% draws when playing as X.                                                  |
| <b>Our work</b>                 | <b>4×4 Tic-Tac-Toe, 3-in-a-row</b> | <b>Tabular Q-learning + tactical policy + DDA</b>                        | <b>800 training + 200 frozen evaluation games, five seeds</b>         | <b>Full model achieved 95.2% against Random and 55.7% against Heuristic. The DDA mechanism produced an average final difficulty of 2.8 against the Heuristic opponent.</b>                      |

Konen and Bartz-Beielstein [11] and Timóteo et al. [15]. Konen and Bartz-Beielstein evaluated reinforcement learning on a 3×3 Tic-Tac-Toe board using a function-approximation-based TD approach. Their evaluation included both selected-state move accuracy and game tournaments. Their best reported agent, trained through 10,000 self-play games, achieved a 100% win rate when playing against a random opponent as the starting player and achieved only draws against Minimax, which is consistent with the theoretical behavior expected against an optimal opponent [11]. This study is particularly relevant because it demonstrates the importance of carefully selecting the evaluation protocol and the state representation when assessing reinforcement learning in small board games.

The recent study by Timóteo et al. [15] provides another direct Tic-Tac-Toe reference. Their Q-learning system was evaluated on a 3×3 board and reported a 77.38% victory rate against a Random opponent when playing as X and a

32.73% victory rate when playing as O. Against MiniMax, the Q-learning agent achieved 0% victories and a high proportion of draws when playing as X. These findings demonstrate two important properties relevant to the present work. First, performance is strongly affected by the opponent and the agent’s player position. Second, a strong deterministic opponent can cause the evaluation to become dominated by the difficulty of the environment rather than by differences between learning configurations.

The present study differs from both of these works by using a  $4 \times 4$  board with a 3-in-a-row winning condition rather than conventional  $3 \times 3$  Tic-Tac-Toe. Furthermore, the proposed system does not rely exclusively on Q-learning. Instead, it combines tabular Q-learning with explicit tactical rules and dynamic difficulty adjustment. Under the frozen evaluation protocol, the Full configuration achieved a 95.2% mean win rate against the Random opponent and a 55.7% mean win rate against the Heuristic opponent. These results indicate strong performance against weak and moderately tactical opponents, although they should not be interpreted as directly superior to the results of the  $3 \times 3$  studies because of the different game configuration and evaluation protocol.

The DDA-oriented studies provide a different but complementary benchmark. Silva et al. [12] developed a DDA mechanism for MOBA games that dynamically switches opponent behavior according to estimated player performance. Their experiments demonstrated that the adaptive mechanism could maintain balanced matches across several simulated skill levels, including approximately 85% balanced matches in one condition and 80% balance against a hard opponent. These results support the principle that DDA should be evaluated not only through the absolute winning rate of the AI agent, but also through the ability of the system to maintain an appropriate relationship between player and opponent strength.

Climent et al. [13] proposed an RL-based DDA framework for single-player action games using two Q-tables. Their design explicitly targeted an approximately balanced winning/losing ratio and personalized challenges based on player skill. This objective is closely related to the DDA component of the present work. In the proposed Tic-Tac-Toe system, however, difficulty is controlled through a skill estimate derived from recent game outcomes and mapped to five difficulty levels. Against the Heuristic opponent, the Full configuration achieved a 55.7% win rate, with an average final difficulty of approximately 2.8 and an average final skill of 52.2. The result is substantially closer to a balanced interaction than the 95.2% win rate obtained against the Random opponent, indicating that the adaptive mechanism reacts to the increasing challenge posed by a stronger opponent.

Similarly, de Oliveira and Chaimowicz [14] formulated reinforcement learning for DDA as a skill-matching problem. Their approach introduced a reward structure encouraging similar relative skill between the player and opponent and a penalty limiting excessive opponent winning or losing. The authors also investigated regularization to improve generalization to previously unseen opponents. This perspective is important for interpreting the present results: maximizing agent win rate is not necessarily the correct objective for an adaptive game

agent. A system intended for personalized gameplay should instead seek an appropriate challenge level.

The ablation results reinforce this distinction. The No Adaptive configuration achieved a 38.2% win rate against the Heuristic opponent, whereas the Full configuration achieved 55.7%. The No Adaptive agent maintained a fixed difficulty level, while the Full agent adjusted its difficulty according to its estimated player skill. Therefore, the improvement associated with DDA should be interpreted as improved adaptation rather than simply improved competitive strength.

The benchmark analysis also reveals an important limitation of the current evaluation. The Strong Tactical opponent produced a 0% agent win rate for every configuration. Consequently, this opponent acts as a saturated stress test rather than a discriminative benchmark. A future evaluation should include a true optimal or Minimax opponent, following the protocol used in Tic-Tac-Toe studies such as Konen and Bartz-Beielstein [11] and Timóteo et al. [15]. Such an experiment would provide a stronger measure of whether the learned policy can approach optimal non-losing play rather than merely outperform heuristic opponents.

Overall, the comparison demonstrates that the proposed work occupies a different point in the design space from conventional Tic-Tac-Toe reinforcement learning. Prior Tic-Tac-Toe studies primarily emphasize learning game strategies and evaluating win/draw/loss performance, whereas the proposed system additionally incorporates tactical safeguards and player-oriented difficulty adaptation. Consequently, its main contribution is not a claim of superior raw Tic-Tac-Toe playing strength, but the integration of reinforcement learning and dynamic difficulty adjustment within a larger, adaptive game-playing framework.

## 4 Conclusion

This work presented a Hybrid Adaptive Reinforcement Learning Framework for dynamically adjusting game difficulty according to player skill in a  $4 \times 4$  Tic-Tac-Toe environment. The proposed system combines Tabular Q-learning,  $\epsilon$ -greedy exploration, rule-based tactical decision-making, player expertise estimation, and Dynamic Difficulty Adjustment (DDA). Unlike conventional fixed-difficulty approaches, the system continuously evaluates recent player performance and adjusts the behavior of the RL agent across five difficulty levels, ranging from Very Easy to Expert. The framework also uses online learning, allowing gameplay interactions to directly contribute to the learning process without requiring a pre-collected training dataset, while persistent player profiles allow learned Q-values and performance information to be retained across multiple sessions.

The experimental results demonstrate that the proposed components contribute differently to the overall gameplay behavior. The ablation study showed that the tactical policy had the strongest immediate effect on game-playing performance, with its removal reducing the mean win rate against the Heuristic

opponent from **55.7%** to **1.4%**. The adaptive difficulty mechanism also produced a substantial difference against the Heuristic opponent, where the complete model achieved a **55.7%** win rate compared with **38.2%** when adaptive difficulty was removed. Against a random opponent, the full model achieved a **95.2%** win rate, while removing the tactical policy reduced the win rate to **40.7%**. These results indicate that tactical reasoning and adaptive difficulty are important components of the proposed framework. However, the contribution of Q-learning was less evident within the relatively short experimental period, as the rule-based baseline (No Q) achieved a comparable or higher short-term win rate against the Heuristic and Random opponents (83.6% and 98.5%, respectively). This suggests that the benefits of learned state-action values may require longer interaction histories, a larger state space, or opponents whose behavior cannot be fully captured by hand-crafted tactical rules.

Despite these results, the current system has several limitations. The proposed framework was demonstrated using a  $4 \times 4$  Tic-Tac-Toe environment, which is considerably simpler than modern video games with larger and more complex state and action spaces. The player expertise estimation is also based primarily on recent game outcomes, which may not fully capture other aspects of player skill, such as decision quality, response time, or consistency. In addition, the No Adaptive configuration used in the ablation study was implemented to hold the difficulty level fixed rather than allowing it to respond to the player; due to an implementation detail in the ablation harness, this fixed level was Very Easy rather than the intended Normal setting, meaning the No Adaptive baseline used the least tactically active and most exploratory policy available to the agent. Consequently, part of the performance gap between the Full and No Adaptive configurations should be interpreted with this in mind, and a corrected fixed-difficulty baseline is identified as a priority for future revision. Furthermore, the experimental evaluation was conducted over a limited number of episodes, and the Strong Tactical opponent resulted in a 0% win rate for all evaluated configurations, limiting its ability to distinguish between the individual components of the framework.

Future work can extend the proposed approach in several directions. Longer training and evaluation periods can be used to better investigate the contribution of Q-learning and long-term adaptation. More comprehensive player models could incorporate additional gameplay features beyond win/loss outcomes to provide a more accurate estimation of player skill. The framework could also be evaluated in more complex game environments with larger state spaces and richer action sets, potentially using function-approximation or deep reinforcement learning methods. In addition, larger-scale user studies could be conducted to evaluate whether the dynamically adjusted difficulty improves player engagement, satisfaction, and perceived challenge. These extensions would provide a stronger evaluation of the framework and help determine its applicability to more complex adaptive gaming environments.

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